An investigation and examination on flexible behaviours — the case of group chase and escape —

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Abstract

In the real world, we often need to cope with unexpected or unknown events. One of the widely used suggestions for these situations is to deal flexibly (flexible behaviour). However, as the event is unexpected, we cannot know the appropriate action previously. We can know it after the event is settled. Furthermore, it is very interesting why flexible behaviours desired for many cases. Hence, we investigate and examine the essentiality of flexible behaviours using recently proposed model, the group chase and escape model to produce various situations.

Keyword: group chase and escape, flexible behaviour, human society, multi-agent simulation

1 Introduction

In the real world, we are often desired to deal flexibly when we take some actions. We consider two the reasons why flexible behaviours are desired: (i) we need to cope with unexpected or unknown events, and (ii) it is preferable to take appropriate actions. To settle unknown problems or unexpected situations in the real world, our experience is useful in some cases, we need to select one of the prepared countermeasures in other cases, and we need to take unprepared action when prepared countermeasures do not work in another case. To cope with these situations, we consider that flexible behaviours are desired. Although flexible behaviours are very appealing strategy, we consider that it is unobvious and heuristic. Hence, in this study, we attempt to find what the flexible behaviours are using group chase and escape model.

2 Methodology and Approach

Although chase and escape model is an old problem, group chase and escape model is recently proposed by Kamimura and Ohira [1]. The model is simple multi-agent simulation model and agents divided into two group chase or escape each other. Also, the model is an imitation of a game of Tag and is similar to ecological system with predator prey model [1].

In 18th century, mathematicians began to study the issue relating to "chase and escape" on one-to-one situation. Recently, mathematicians study animal, insect and human being on group behaviour using computer simulation. Group chase and escape includes two aspects, group with a sense of purpose and chase and escape.

In group chase and escape model, the basic behaviour rule is simple. There are two types of agents in the model. One is chaser and the other is target. Chasers approach the nearest targets and targets escape from the nearest chasers. Each agent moves based on this simple rule. However, we note that group behaviour of agents is complex because of agents' interplay or mutual action. The features of group chase and escape is as follows: (1) the time it takes chasers to catch all targets (life time) (2) the relation between agent density and efficiency of life time (3) the structure between chaser and target emerged on simulating and so on.

3 Summary

In human society we are required flexible behaviour. However, we consider that the essentiality and methodology is not established. In this study, using group chase and escape model, we attempt to know the essentiality of flexible behaviours.

References

[1] Atsushi Kamimura and Toru Ohira, "Gorup chase and Escape", New Journal of Physics 12, 053013 (2010).